Lecture 26

Hardy space and the Khalfin obstruction

Objectives. *Positive-time square-integrable data give holomorphy in the upper half-plane. A different support condition, a lower bound on a spectral variable, obstructs exponential decay of a scalar Fourier integral. We prove this obstruction for a nonzero nonnegative integrable density. Applying it to a Hamiltonian requires a cited spectral representation and an explicit absolute-continuity assumption.*

By the end of this lecture you will be able to:

- Construct a Hardy-space function from positive-time square-integrable data.
- State the precise cited converse with its norm and boundary values.
- Prove the scalar Khalfin obstruction by analytic continuation in a strip.
- Identify the spectral-density hypothesis needed for the quantum corollary.

26.1 Positive-time data and Hardy space

The *Hardy space* $H^2(\mathbb{H})$ consists of holomorphic functions h on $\mathbb{H} = \{z : \operatorname{Im} z > 0\}$ for which the following norm is finite:

$$\|h\|_{H^2}^2 = \sup_{y>0} \int_{\mathbb{R}} |h(x+\mathrm{i}y)|^2 \,\mathrm{d}x. \tag{26.1}$$

The variable y in this definition is the height in the upper half-plane. It is unrelated to the dependent variable of an ODE.

For $g \in L^2(0,\infty)$ define

$$h(z) = \frac{1}{\sqrt{2\pi}} \int_0^\infty g(t) e^{\mathrm{i}zt} \,\mathrm{d}t, \qquad z \in \mathbb{H}. \tag{26.2}$$

The positive exponent is the time–frequency convention of Equation (12.5). It is opposite to the negative exponent of the spatial Fourier transform. For $\operatorname{Im} z = y > 0$, Cauchy–Schwarz bounds the absolute integral by $\|g\|_2 \, (\int_0^\infty e^{-2yt} \,\mathrm{d}t)^{1/2}$. On each compact subset of $\mathbb{H}$ the same argument bounds the integrand and all its z derivatives using $t^n e^{-y_0 t}$ for some $y_0 > 0$. The holomorphic parameter theorem, Theorem 10.9, proves holomorphy.

Extend g by zero to negative t. At height y, the function $h(x+\mathrm{i}y)$ is the inverse unitary Fourier transform of $g(t)e^{-yt}$. Plancherel, Theorem 7.7, therefore gives

$$\int_{\mathbb{R}} |h(x+\mathrm{i}y)|^2 \,\mathrm{d}x = \int_0^\infty |g(t)|^2 e^{-2yt} \,\mathrm{d}t, \qquad \|h\|_{H^2} = \|g\|_2 \,. \tag{26.3}$$

The supremum is the limit as $y \downarrow 0$, by dominated convergence. Also $g(t)e^{-yt} \to g(t)$ in L^2. Plancherel shows that $h(x+\mathrm{i}y)$ has an L^2 limit on the boundary. An L^2 boundary value here means convergence in norm. It does not assert convergence at every boundary point.

For example, take $g(t) = e^{-at}$ with $a > 0$. Then

$$h(z) = \frac{1}{\sqrt{2\pi}(a - \mathrm{i}z)} = \frac{\mathrm{i}}{\sqrt{2\pi}(z + \mathrm{i}a)}, \qquad \|h\|_{H^2}^2 = \frac{1}{2a}.$$

The pole is below the real axis, as positive-time support requires. At height y the squared norm is $1/(2(a+y))$, so its supremum is $1/(2a)$.

26.2 The cited converse

Theorem 26.1 (Half-plane Paley–Wiener theorem · *cited*)**.** *Every $h \in H^2(\mathbb{H})$ has a unique representation Equation (26.2) by a function $g \in L^2(0,\infty)$. It has the L^2 boundary value given by the inverse Fourier transform of the extension of g by zero, and the norm identity Equation (26.3) holds.*

The converse is Stein–Shakarchi, *Real Analysis*, §5.2, Theorem 2.1, pp. 213–216. Write h_0 for the boundary value when quoting their theorem. Their exponent is $2\pi\mathrm{i}\xi z$. Substitution $t = 2\pi\xi$ converts their inverse transform to ours, with $g(t) = \widehat{h_0}(t/(2\pi))/\sqrt{2\pi}$ in their normalisation. This also converts their isometry to the norm used here. The construction above proves the forward direction without this converse.

Support and half-plane analyticity must be paired with a growth or norm condition. Holomorphy alone does not imply positive-time support. For example, a nonzero constant is holomorphic on $\mathbb{H}$ but its boundary function is not square-integrable, so it does not belong to this theorem.

26.3 An integrable Fourier inversion lemma

The scalar obstruction needs L^1 inversion rather than the Hardy converse. Recall that for $f \in L^1(\mathbb{R})$ the unitary transform is $\widehat{f}(k) = (2\pi)^{-1/2} \int f(\lambda)e^{-\mathrm{i}k\lambda}\,\mathrm{d}\lambda$.

Lemma 26.2. *If $f \in L^1(\mathbb{R})$ and $\widehat{f} \in L^1(\mathbb{R})$, then*

$$f(\lambda) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} \widehat{f}(k)e^{\mathrm{i}k\lambda}\,\mathrm{d}k$$

for almost every $\lambda \in \mathbb{R}$. The right side is a continuous function of λ.

Proof. Multiply the integrand by $e^{-\varepsilon k^2/2}$. The Gaussian regularisation identity, Lemma 7.4, makes this integral $f * H_\varepsilon$, where H_ε is the mass-one Gaussian. The identity was proved by absolute Fubini for every L^1 input. Since $\widehat{f}$ is integrable, removing the factor by dominated convergence gives uniform convergence in λ to the displayed inverse integral: the error is bounded by $(2\pi)^{-1/2} \int |\widehat{f}(k)||e^{-\varepsilon k^2/2} - 1|\,\mathrm{d}k$. The inverse integral is continuous, again by dominated convergence.

The approximate-identity theorem, Theorem 5.8, gives $f * H_\varepsilon \to f$ in L^1. Choose a sequence $\varepsilon_n \downarrow 0$ with $\|f * H_{\varepsilon_n} - f\|_1 \le 2^{-n}$. Tonelli makes the sum of these absolute errors finite almost everywhere. Thus the sequence also converges pointwise to f almost everywhere. Its uniform limit is the inverse integral, proving the assertion. ∎

The conclusion identifies an almost-everywhere representative with a continuous function. It does not make every originally assigned point value of an arbitrary L^1 function continuous.

26.4 The scalar Khalfin obstruction

Fix $\hbar > 0$. Let $\rho \in L^1(\mathbb{R})$ be nonnegative, nonzero, and supported in $[E_0, \infty)$ for some $E_0 \in \mathbb{R}$. Support here means that $\rho(\lambda) = 0$ almost everywhere for $\lambda < E_0$. Define the scalar amplitude by

$$A(t) = \int_{\mathbb{R}} \rho(\lambda) e^{-\mathrm{i}\lambda t/\hbar}\, \mathrm{d}\lambda, \qquad t \in \mathbb{R}. \tag{26.4}$$

If $\int \rho = 1$, its squared modulus is a normalised survival probability. The proof needs only the stated L^1 density hypotheses.

Theorem 26.3 (Scalar Khalfin obstruction). *There do not exist $\alpha > 0$, $C > 0$ and $t_1 \geq 0$ such that $|A(t)| \leq Ce^{-\alpha t}$ for every $t \geq t_1$.*

Proof. Suppose such constants exist. Since ρ is real, $A(-t) = \overline{A(t)}$. Also $|A(t)| \leq \|\rho\|_1$ for all t. Increasing C to cover the compact interval $[-t_1, t_1]$ therefore gives $|A(t)| \leq C' e^{-\alpha|t|}$ on the whole real line. In particular $A \in L^1(\mathbb{R})$.

Define the analytic density candidate in a strip by

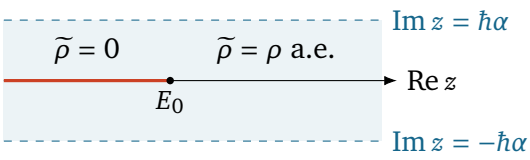
$$\widetilde{\rho}(z) = \frac{1}{2\pi\hbar} \int_{\mathbb{R}} A(t) e^{\mathrm{i}zt/\hbar}\, \mathrm{d}t, \qquad |\operatorname{Im} z| < \hbar\alpha. \tag{26.5}$$

On each compact subset of this strip the integrand is bounded by $C' e^{-d|t|}$ for some $d > 0$. The parameter theorem, Theorem 10.9, therefore makes $\widetilde{\rho}$ holomorphic there. Since $A(t) = \sqrt{2\pi}\widehat{\rho}(t/\hbar)$, the change of variable $t = \hbar k$ and Lemma 26.2 give $\widetilde{\rho}(\lambda) = \rho(\lambda)$ for almost every real λ.

The continuous function $\widetilde{\rho}$ thus vanishes almost everywhere on $(-\infty, E_0)$. It vanishes at every point of that interval: a nonzero value would, by continuity, persist on a real subinterval of positive measure. The identity theorem, Theorem 10.6, makes $\widetilde{\rho}$ identically zero on the connected strip. Its almost-everywhere equality with ρ now says $\rho = 0$ almost everywhere, contrary to the hypothesis. ∎

The contradiction uses each hypothesis for a separate step. Reality of the density turns a future-time bound into a bound in both time directions. The density is integrable by hypothesis. The assumed bound, extended to both time directions, makes the amplitude integrable too. These two facts permit Fourier inversion. The lower support bound supplies a whole zero interval for the continuous analytic representative. The identity theorem then forces that representative to vanish. None of these steps uses the onto Hardy-space theorem.

The lower support bound and the strip have different variables.



Exponential decay in time would extend the spectral density to this open strip. The density labels refer to the real axis; the almost-everywhere equality holds on the whole real axis. Its zero interval would then force the whole extension to vanish.

For example, take $\rho(\lambda) = 2e^{-2\lambda}\mathbb{1}_{(0,\infty)}(\lambda)$. Direct integration gives

$$A(t) = \frac{2}{2 + \mathrm{i}t/\hbar}, \qquad |A(t)|^2 = \frac{4}{4 + (t/\hbar)^2}.$$

This amplitude has an algebraic tail. The theorem does not prescribe one universal power law, or a pointwise polynomial lower bound for every density. It excludes an eventual exponential upper bound.

26.5 The conditional quantum interpretation

For a self-adjoint Hamiltonian H and a normalised vector ψ, the spectral theorem gives a probability measure μ_ψ such that

$$\left\langle \psi, e^{-\mathrm{i}Ht/\hbar}\psi \right\rangle = \int_{\mathbb{R}} e^{-\mathrm{i}\lambda t/\hbar}\, \mathrm{d}\mu_\psi(\lambda).$$

This Hamiltonian-to-measure statement is cited from Teschl, Ch. 3. For unbounded Hamiltonians its functional calculus is outside this course. If H is bounded below by E_0, the measure is supported in $[E_0, \infty)$. It need not have a density. An eigenvector, for example, gives a point mass.

Corollary 26.4 (Quantum consequence · *sketched*)**.** *Suppose the cited spectral representation applies and* $\mathrm{d}\mu_\psi(\lambda) = \rho(\lambda)\,\mathrm{d}\lambda$ *with a nonnegative* L^1 *density supported in a half-line. Then the survival amplitude has no eventual exponential upper bound with positive rate.*

The scalar proof is complete. The tag records the cited spectral bridge. The assertion here does not extend that proof to arbitrary singular measures. A resonance approximation may describe an intermediate time interval without giving the forbidden bound for every sufficiently large time.

Summary.

- Positive-time L^2 data give an upper-half-plane Hardy function isometrically. The onto Paley–Wiener converse is cited.
- Integrable Fourier data with integrable transform satisfy inversion almost everywhere.
- A nonzero nonnegative L^1 spectral density supported in a half-line cannot have an eventually exponentially bounded amplitude.
- The quantum corollary requires both the cited spectral bridge and an absolutely continuous scalar spectral measure.

The half-plane conventions will also fix the factors in Lecture 27.

References

E. M. Stein and R. Shakarchi, *Real Analysis*, Princeton Lectures in Analysis III, Princeton, 2005, §5.2, Theorem 2.1, pp. 213–216. G. Teschl, *Mathematical Methods in Quantum Mechanics*, online version 12 February 2009, §3.1, pp. 89–95, for scalar spectral measures and bounded-Borel functional calculus. The Fourier inversion and scalar strip argument above use the proved results of Part I rather than a cited singular-measure version of Khalfin's theorem.